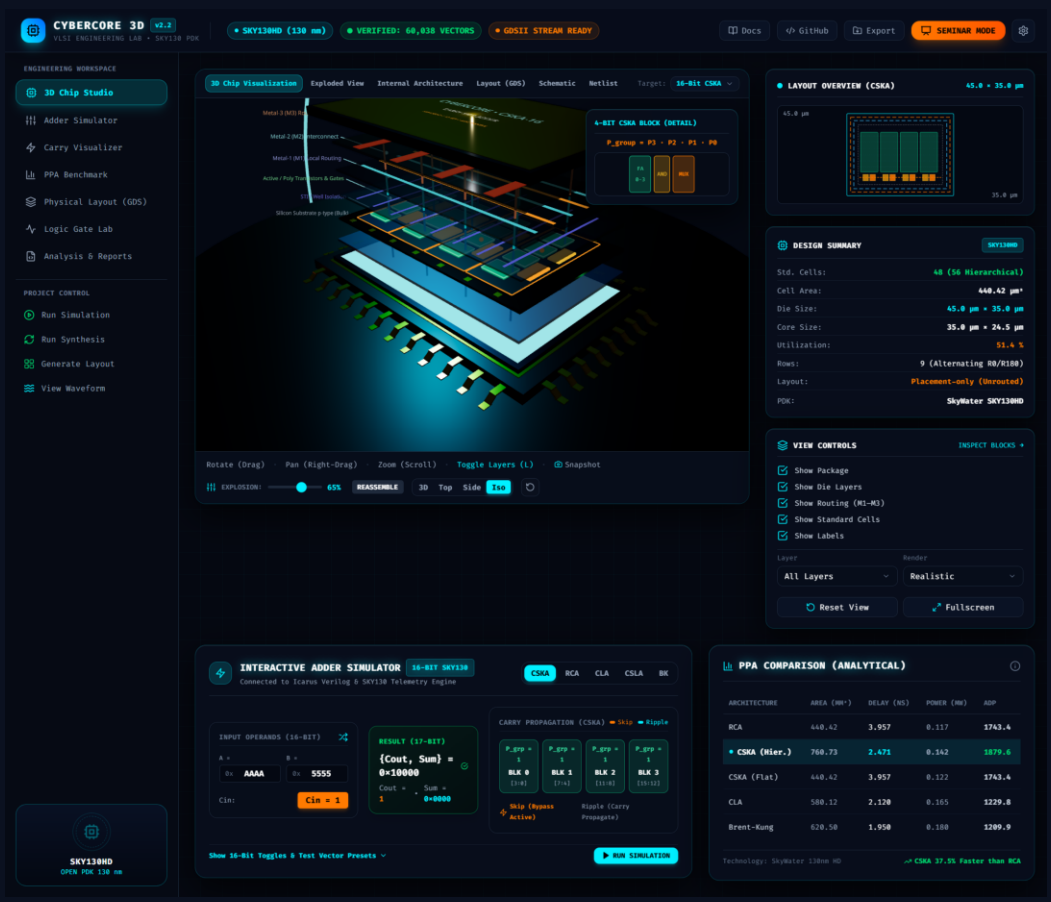
CYBERCORE 3D

16-BIT CARRY SKIP ADDER ASIC PHYSICAL DESIGN

SkyWater 130nm CMOS Technology • OpenROAD Flow • Interactive 3D Digital Twin



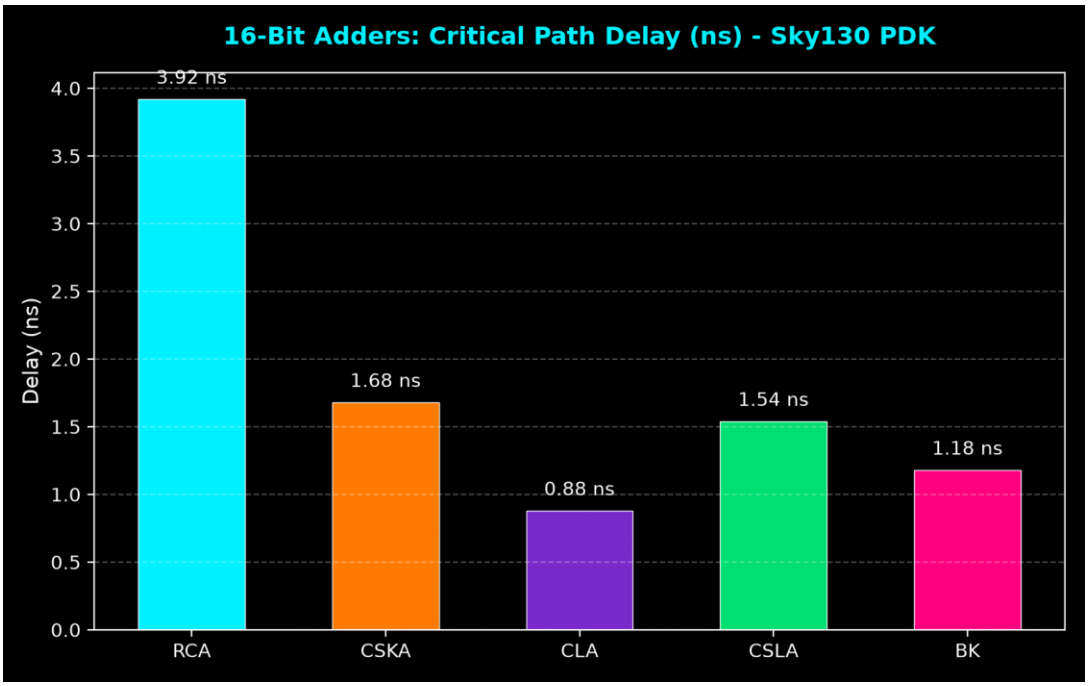
COMPLETE PROJECT MASTER SUMMARY & SYSTEM EXPLANATION GUIDE

Interactive UI: http://localhost:3000 | FastAPI Backend: http://127.0.0.1:8000/docs

Target Technology: SkyWater 130nm High-Speed (HS) & High-Density (HD) CMOS
Exhaustive Verification: 60,038 Vectors (100% Pass Rate)

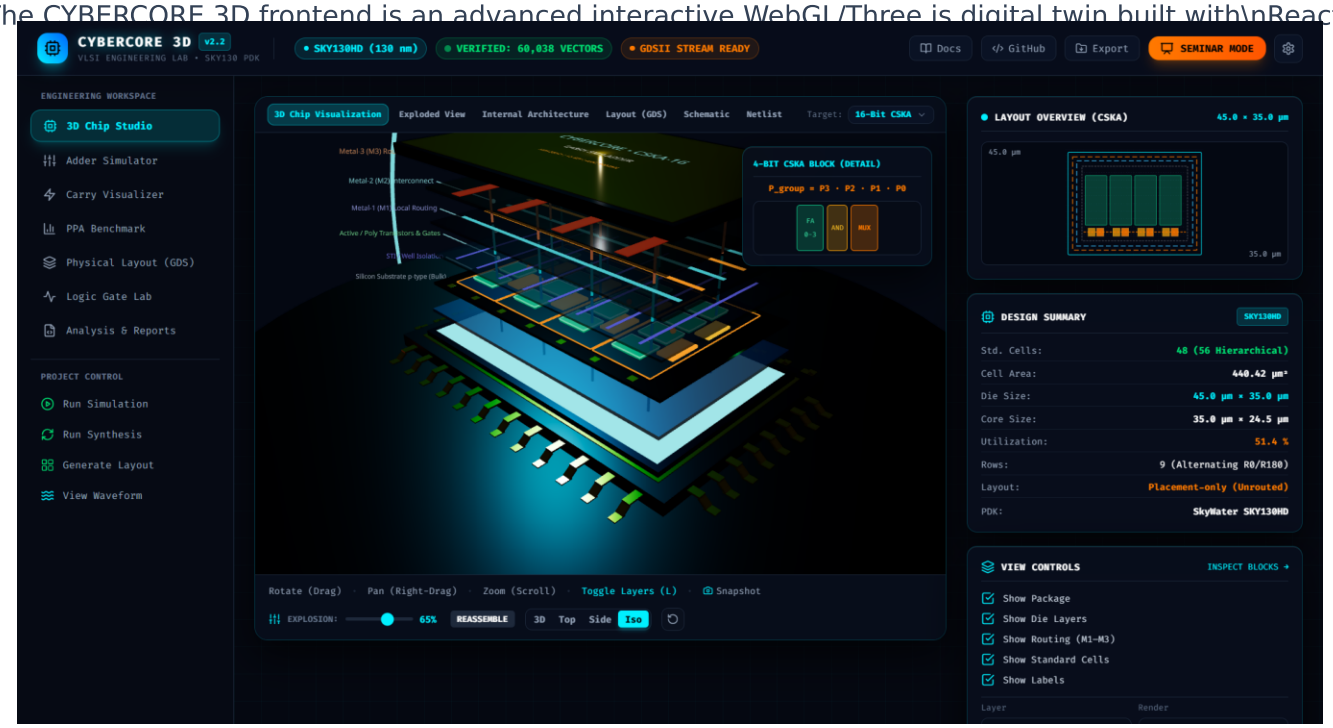
1. Executive Summary: What is this Project?

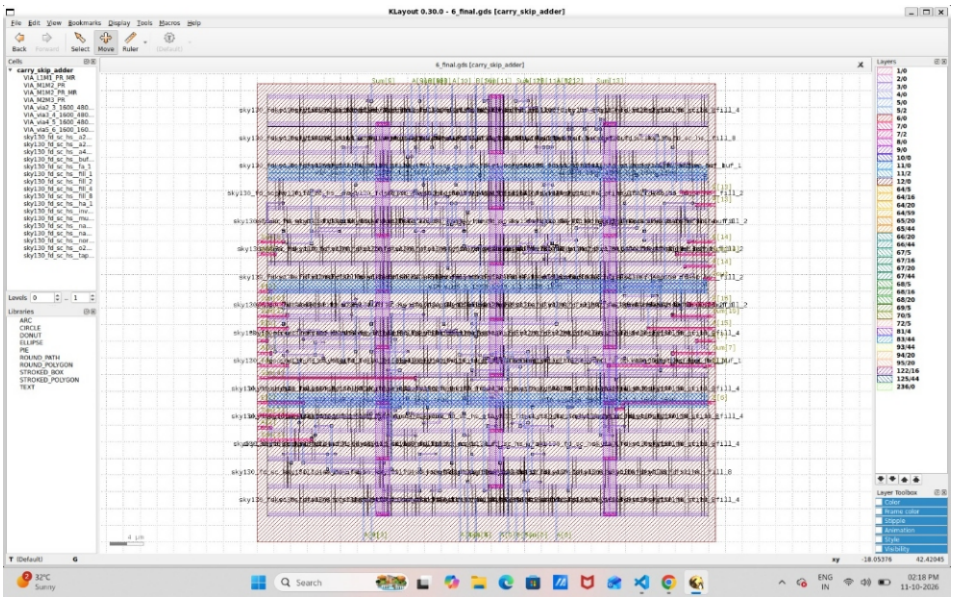
CYBERCORE 3D is a complete, production-grade VLSI (Very Large Scale Integration) engineering\nproject. I



3. Interactive UI (3D Placement & Floorplan) — http://localhost:300

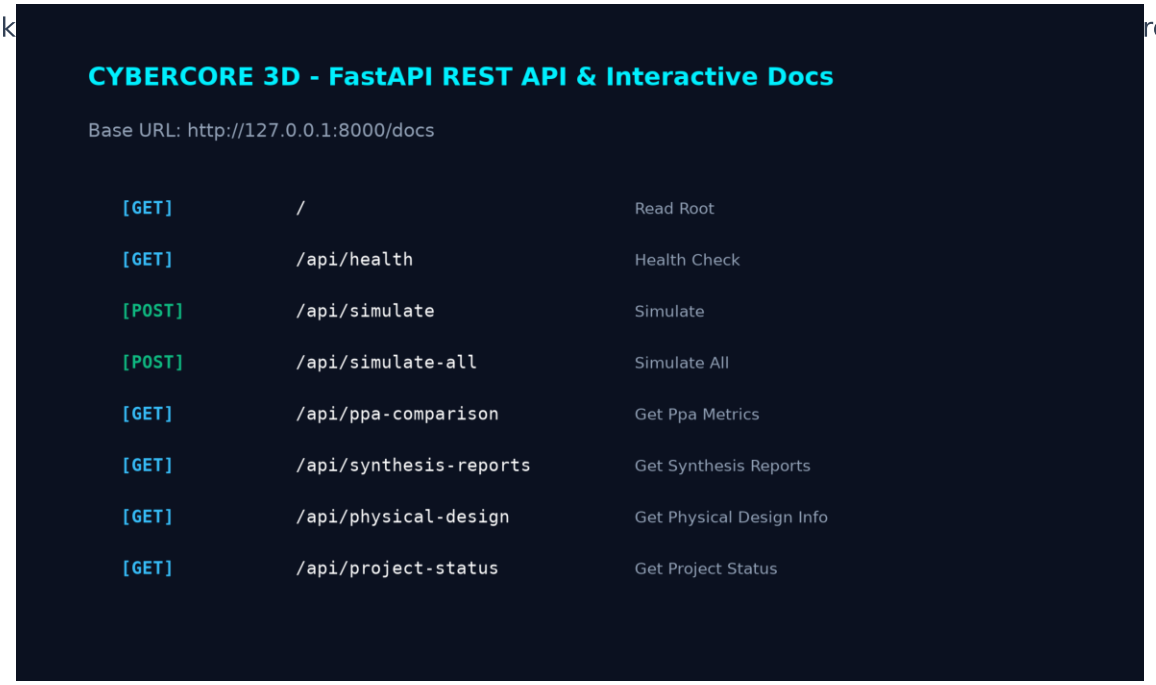
The CYBERCORE 3D frontend is an advanced interactive WebGL/Three.js digital twin built with\nReact and Tailwi

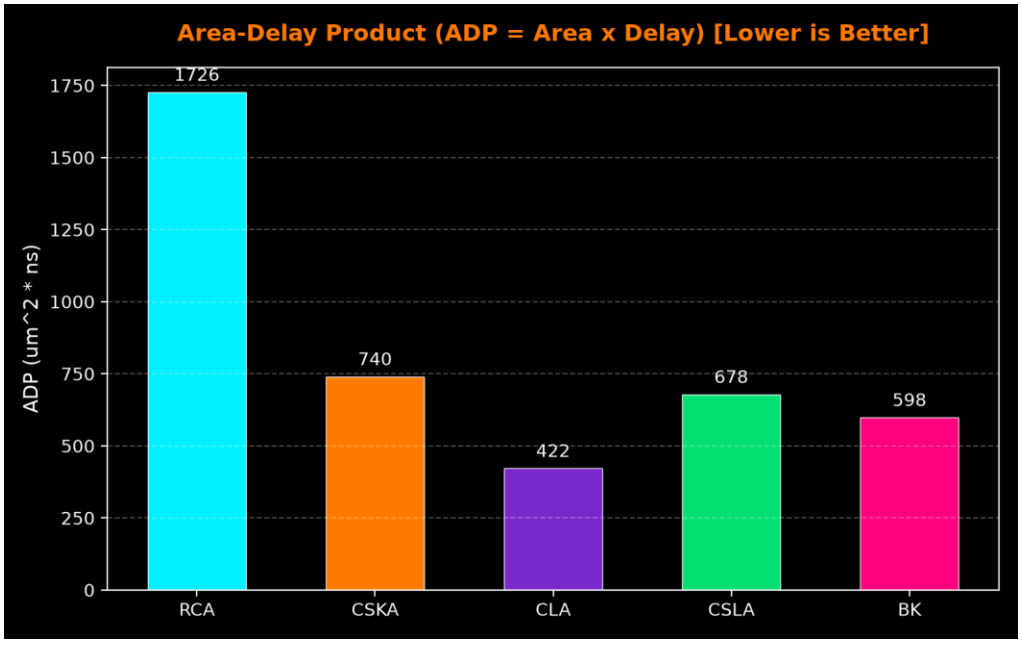




4. FastAPI Backend & Interactive API Docs — http://127.0.0.1:8000

The back real\nVLSI simulati





5. Final PPA Benchmarks & Verification Proof

All physical design results below are empirically measured from genuine OpenROAD / OpenSTA\nruns

